

MAMBA-NIKE Optimized Implementation x86 Self-Evaluation Report

MAMBA-NIKE Submission Team

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Document Information

Algorithm	MAMBA-NIKE
Implementation track	AVX2 Optimized
Target architecture	x86-64
Implementation directory	API_PKC/Implementations/Optimized_Implementation/
Parameter authority	parameters.json
Parameter SHA-256	a34156a3dcc6c80328e5bf8814d7c1aa4764dd1ac8e257c455000e124194ffdd
Evaluation code	ci/optimized_self_evaluation.py
Machine-readable evidence	ci/optimized_self_evaluation_data.json
Tabular evidence	ci/optimized_self_evaluation_data.csv

1 Scope

This report evaluates only the submitted AVX2 Optimized implementation. It does not combine evidence from the other implementation track. The evaluated profiles are MAMBA-NIKE-128, MAMBA-NIKE-192, MAMBA-NIKE-256, MAMBA-NIKE-384, and MAMBA-NIKE-512. The evaluation covers dependencies, the frozen parameter record, implementation-boundary enforcement, canonical known-answer tests, protocol agreement observed by the benchmark harness, and x86 cycle measurements.

The scheme operates in

$$R_q = \mathbb{Z}_q[X]/(X^n + 1), \quad q = 8192 = 2^{13},$$

and uses SHAKE256 for transcript-bound shared-secret derivation. No Cortex-M4 or Additional Implementation material is part of this report.

2 Dependencies

- Python 3.9 or newer; only the Python standard library is used.
- GNU Make and Bash.
- A GCC-compatible C99 compiler with AVX2 support.
- An x86-64 processor reporting AVX2 support.
- No third-party cryptographic library is required.

The evaluator itself has no network dependency and does not install packages. The optimized evaluator stops with a failure when AVX2 is unavailable instead of silently reporting portable fallback results as optimized results.

3 Implementation Boundary

The evaluated implementation preserves the same API, parameter constants, serialization, protocol flow, and KAT outputs as the Reference implementation. When AVX2 is enabled, products with one centered-binomial operand use `poly_mul_small()` with fixed loop counts, fixed memory accesses, 16-bit SIMD lanes, lane-wise accumulation, and one final reduction modulo $q = 2^{13}$. The optimized build also links the submitted AVX2 ChaCha20 assembly. Toom–Cook-4 remains the non-AVX2 fallback.

- The source-boundary scan requires AVX2 intrinsics, the `__AVX2__` guarded path, `poly_mul_small()`, and `chacha.S`.
- Every profile must retain `toom4_mul()` as the portable fallback.
- AVX2 is forced during KAT reproduction with `AVX2=1`.
- KAT binaries are built in temporary directories so the submitted implementation tree is not polluted with build products.

4 Self-Evaluation Procedure

The implementation-specific runner performs the following ordered checks:

1. verify required tools and processor capabilities;
2. verify the immutable `parameters.json` SHA-256 value;
3. scan the selected implementation tree for the documented implementation boundary;
4. copy each profile to a temporary directory;
5. build and run its KAT executable;
6. compare the generated KAT file byte for byte with `API_PKC/Test_Vectors/`;
7. run the submitted 1,000-iteration RDTSC benchmark;
8. verify zero observed agreement failures;
9. write JSON and CSV evidence; and
10. verify that `parameters.json` was unchanged.

```
# Run from the repository root.
python3 ci/optimized_self_evaluation.py
# Structural and KAT checks without the 1,000-run benchmark:
python3 ci/optimized_self_evaluation.py --skip-benchmark
```

5 Known-Answer Test Results

All five generated KAT files matched the canonical Reference test-vector files byte for byte. The digest prefix below is included for convenient evidence identification; the complete digest is stored in the JSON evidence file.

Table 1: AVX2 Optimized KAT reproduction results.

Profile	File bytes	Result	SHA-256 prefix
MAMBA-NIKE-128	213180	PASS	85ea2c7e0762ca3e...
MAMBA-NIKE-192	223420	PASS	965a521ebbc5fd6d...
MAMBA-NIKE-256	223420	PASS	cb420026a0962881...
MAMBA-NIKE-384	464700	PASS	eb125000e598fbca...
MAMBA-NIKE-512	465340	PASS	a340e7bfd2002220...

6 x86 Performance and Agreement Data

The submitted benchmark harness uses RDTSC and reports the arithmetic mean over 1,000 timed executions after warm-up. KeyGen, responder Pass1, and initiator Derive are measured separately. The same 1,000 executions also check shared secret agreement.

Table 2: AVX2 Optimized results (kCycles; 1,000 executions).

Profile	n	PK B	M_1 B	KeyGen	Pass1	Derive	Agree / Fail
MAMBA-NIKE-128	1024	1184	1568	151	293	113	1000 / 0
MAMBA-NIKE-192	1024	1312	1568	154	270	115	1000 / 0
MAMBA-NIKE-256	1024	1312	1568	177	325	115	1000 / 0
MAMBA-NIKE-384	2048	2848	3360	349	683	322	1000 / 0
MAMBA-NIKE-512	2048	2848	3360	343	690	317	1000 / 0

All recorded benchmark exchanges agreed and the observed failure count was zero. This empirical result is not presented as a cryptographic upper bound on the decryption or reconciliation failure probability.

7 Data Format

The JSON evidence records environment metadata, dependency status, the implementation boundary, the parameter hash before and after evaluation, full KAT digests, timing data, agreement counts, and provenance notes. The CSV file contains one normalized benchmark row per submitted profile. Both files are plain UTF-8 and can be consumed without a project-specific parser.

8 Security and Correctness Limitations

This implementation self-evaluation verifies reproducibility and consistency of the submitted code. It does not replace independent cryptanalysis, formal verification, constant-time leakage testing, or a certified numerical failure probability analysis. The parameter manifest and every profile-specific `params.h` remain frozen and are not rewritten by the evaluator.

9 Conclusion

The AVX2 Optimized implementation passed the implementation-specific source boundary checks and reproduced all five canonical KAT files. The associated 1,000-run x86 benchmark dataset recorded zero shared-secret disagreements. The evaluation code, JSON evidence, and CSV evidence are separated from the other implementation track so that the submitted Reference and Optimized implementations can be assessed independently.